
SmartGridBench: MCP-Based Industrial Agent Benchmarking for Smart Grid Transformer Operations

Anonymous Author(s)

Affiliation

Address

email

Abstract

Industrial LLM agents are increasingly evaluated as tool-using systems, but current benchmarks underrepresent maintenance workflows where decisions require coordinated reasoning across diverse operational interfaces. We present SmartGridBench, an AssetOpsBench extension for smart-grid power transformer operations. SmartGridBench pairs a license-clean synthetic transformer fleet with 36 validated transformer-maintenance scenarios, four tool domains, and both direct-call and Model Context Protocol (MCP) tool surfaces. The benchmark separates two often-confounded factors in industrial-agent evaluation: protocol transport overhead and orchestration strategy. In controlled A100 runs with Llama-3.1-8B-Instruct, we evaluate direct Agent-as-Tool, MCP Agent-as-Tool, optimized MCP transport, Plan-Execute, Verified Plan-Execute, Self-Ask, optimized-serving follow-ons, and mitigation ablations over repeated judged trials. The current evidence floor contains 2,420 judged trajectories across 37 row groups, including a 31-scenario core matrix and hosted-70B top-ups. The resulting matrix shows that MCP standardization adds measurable latency without a corresponding quality gain, that PE-family orchestration raises judged quality at higher latency, and that Self-Ask and optimized transport are not uniformly positive. A mitigation ladder over evidence guard, repair, and fault-risk adjudication makes failure accounting more auditable: repair is the narrowest positive intervention, while guard and adjudication often expose quality-completion trade-offs rather than improving pass rate. SmartGridBench contributes a reusable industrial benchmark resource and an auditable systems study of protocol and orchestration choices in tool-using agents.

1 Introduction

Industrial-agent benchmarks still under-cover Smart Grid transformer diagnostics and maintenance workflows, even though transformer operations are a natural fit for multi-tool agents. A credible maintenance answer may need to retrieve telemetry, interpret dissolved-gas-analysis evidence, forecast degradation, and create or prioritize a work order. These tasks also expose a systems question that many agent evaluations blur: is a result caused by the tool protocol, by the orchestration strategy, by the model, or by failures in evidence accounting?

SmartGridBench addresses this gap by extending AssetOpsBench Patel et al. [2025] with a Smart Grid transformer-maintenance lane and by treating protocol, orchestration, and evidence accounting as benchmark variables. The benchmark is not designed as a model leaderboard. It is an evaluation resource for studying how tool-using industrial agents behave when the tool surface, artifact schema, and judged outputs are kept auditable across cells.

This distinction matters because enterprise deployment practice and benchmark performance do not necessarily reward the same design. Agent-as-Tool often looks strong because iterative tool use creates an implicit reflection loop. Plan-Execute remains attractive in settings that value inspectable plans and handoff structure. MCP standardization Anthropic [2024] is useful for tool interoperability, but it can also introduce measurable runtime overhead. We therefore separate transport from orchestration and reserve reasoning enhancements such as Self-Ask and verification for explicit follow-on rows rather than silently moving the baseline.

Contributions. SmartGridBench contributes:

1. a Smart Grid transformer-maintenance extension to AssetOpsBench with MCP-backed IoT, FMSR, TSFM, and WO tool domains;
2. a validated 36-scenario corpus over a public-safe synthetic transformer fleet;
3. a two-axis evaluation design separating transport overhead from orchestration behavior; and
4. an evidence-backed failure-taxonomy and mitigation analysis for tool-using industrial agents.

2 Benchmark Extension

The benchmark extension is built from public transformer-related data sources Katser and Kozitsin [2021], bantipatel20 [n.d.], easonlai [n.d.], sreshta140 [n.d.], ziya07 [n.d.] reconciled onto a shared synthetic `transformer_id` key. This lets the four tool domains operate over the same asset within one trajectory: an agent can inspect telemetry, diagnose fault evidence, estimate degradation risk, and recommend maintenance actions for a single transformer rather than solving isolated one-tool subtasks.

Table 1: SmartGridBench tool domains. The four-domain shape mirrors AssetOpsBench while changing the asset class to transformer operations.

Domain	Role in a Smart Grid maintenance trajectory
IoT	Retrieve transformer telemetry, operating context, and asset metadata.
FMSR	Interpret dissolved-gas-analysis and failure-mode evidence.
TSFM	Forecast degradation signals and anomaly trajectories.
WO	Create, retrieve, and prioritize maintenance work orders.

Scenario construction follows a realism-first rule: every task names a transformer, preserves cross-tool consistency through the shared asset key, and stays inside documented operational bounds for DGA interpretation, loading, urgency, and work-order fields. The canonical submission corpus contains 36 validated scenario files and five negative schema fixtures. Thirty-one scenarios are hand-authored; five generated candidates were promoted only after manual disposition, bounded edits, and structural validation; the same final disposition pass rejected 10 of 15 generated candidates as near-duplicates. This keeps the generation story conservative: generated scenarios extend the corpus, but they do not replace domain review.

The DGA data pipeline has an additional limitation. Individual scenario fixture values are checked against IEC/IEEE-style operational ranges International Electrotechnical Commission [2022], IEEE Power and Energy Society [2019], Duval and de Pablo [2001], but a separate L3 statistical realism validator finds that the small synthetic DGA distribution does not yet match a real labeled DGA reference distribution bantipatel20 [n.d.], Arias and Mejia Lara [2020]. We therefore claim plausible fixture values and auditable scenario construction, not a real-fleet DGA distribution. This limitation is intentional and important: scenario realism, generated-scenario circularity, and failure accounting are first-class benchmark artifacts rather than post-hoc notes.

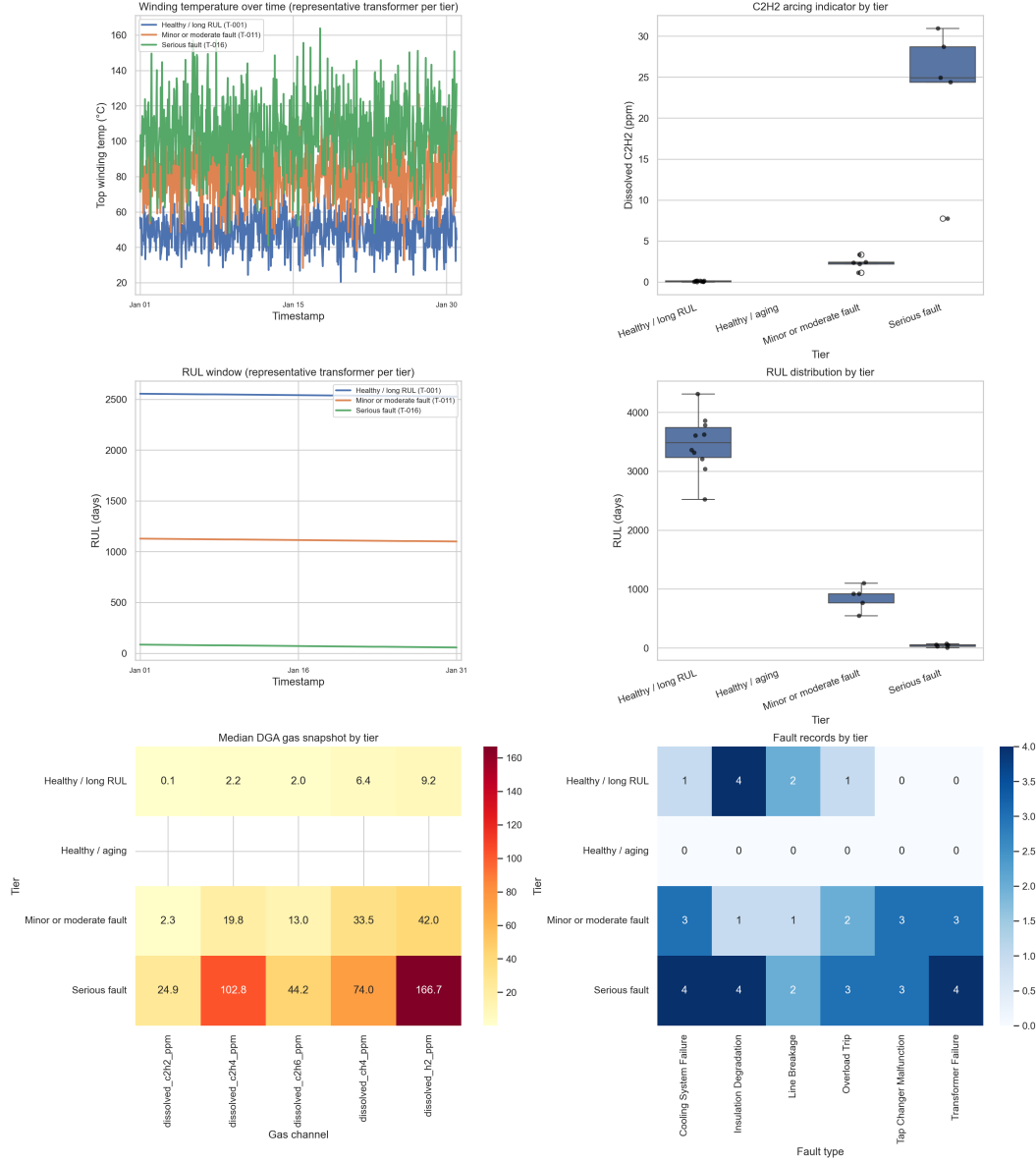


Figure 1: Dataset and asset overview used to build the public-safe synthetic transformer fleet. Source: results/figures/notebook01_dataset_overview.png.

3 System Design

The system keeps one benchmark-facing artifact contract across conditions. scripts/run_experiment.sh writes cell-level config.json and summary.json plus run-scoped raw artifacts under benchmarks/cell_<X>/raw/<run-id>/. Each trial stores raw trajectories, latencies, judge records, run metadata, and analysis-ready CSV exports. This shared layout lets transport and orchestration experiments reuse the same analysis surfaces, Weights & Biases joins, and profiling references instead of inventing per-method logging formats.

Scenario query → Benchmark runner → A/B/C/Y/Z execution
Direct tools or MCP servers → summary.json + latencies.jsonl + raw traces
→ judge outputs + metrics CSVs + failure taxonomy

Figure 2: Shared artifact contract across transport and orchestration cells.

The runtime stack intentionally separates benchmark plumbing from orchestration logic. Plan-Execute, PE + Self-Ask, Verified PE, and mitigation variants all write into the same artifact layout, while the Agent-as-Tool lane uses a shared runner for direct Python tools and MCP servers. This split keeps logging and reproducibility stable even as orchestration implementations evolve.

4 Experimental Design

Experiment 1 varies transport while holding the orchestration style fixed at Agent-as-Tool: Cell A uses direct Python tools, Cell B uses MCP baseline servers, and Cell C uses optimized MCP transport. Cell D is an additional optimized-serving follow-on. Experiment 2 varies orchestration while holding the tool protocol fixed at MCP: Cell B is the Agent-as-Tool anchor, Cell Y is Plan-Execute, and Cell Z is Verified Plan-Execute. PE + Self-Ask and Verified PE + Self-Ask are treated as follow-on rows rather than replacements for the baselines.

Table 2: Core experimental cells. B is the shared anchor: MCP transport baseline in Experiment 1 and Agent-as-Tool orchestration baseline in Experiment 2.

Cell	Display	Axis	Meaning
A	AT-I	Transport	Agent-as-Tool with direct in-process tools.
B	AT-M	Shared anchor	Agent-as-Tool with MCP baseline servers.
C	AT-TP	Transport	Agent-as-Tool with optimized MCP transport.
D	AT-TPQ	Follow-on	Optimized MCP plus optimized serving stack.
Y	PE-M	Orchestration	Plan-Execute with MCP baseline servers.
Z	V-M	Orchestration	Verified Plan-Execute with MCP baseline servers.
YS	PE-S-M	Follow-on	Plan-Execute + Self-Ask.
ZS	V-S-M	Follow-on	Verified Plan-Execute + Self-Ask.

The evidence floor is intentionally tiered. The broad quality table uses the 31-scenario core matrix with five trials per scenario per cell. Latency claims use a tighter six-scenario final A100 cohort where all compared rows have matched latency artifacts. Hosted-70B rows are reported as top-ups and must be read with their own per-cell sample sizes: A/B/C use 45 judged trials each, while Y/YS/Z/ZS use 75 judged trials each. We do not mix these rows into the main 8B table without disclosing sample size.

All paper tables are gated by `results/metrics/evidence_registry.csv` (`include_in_paper=true`), which selects 2,420 paper-grade judged trajectories from a larger 3,716-row evaluation pool and preserves superseded, diagnostic, and invalid-tooling rows for auditability.

One implementation caveat is also disclosed for parity. Most rows use a shared Python 3.11 orchestration environment; vanilla Cell Y uses the upstream AssetOpsBench Plan-Execute CLI under Python 3.12 so that it remains the published-AOB baseline rather than a repo-local compatibility wrapper. This difference is confined to the orchestration client; model serving, tool servers, scenario slices, and decoding settings are held fixed.

103 5 Results

104 5.1 Core 31-scenario quality matrix

Table 3: Post-freeze 31-scenario core matrix. Each row is 31 scenarios times 5 trials, all judged with Llama-3.1-8B-Instruct AI [2024]. Source: `results/metrics/gcp_post175_core31_summary.csv`.

Cell	Meaning	Judged trials	Judge mean	Judge pass
A	Agent-as-Tool direct	155	0.474	67/155 (43.2%)
B	Agent-as-Tool MCP	155	0.442	57/155 (36.8%)
C	Agent-as-Tool optimized MCP	155	0.473	61/155 (39.4%)
Y	Plan-Execute MCP	155	0.582	76/155 (49.0%)
YS	Plan-Execute + Self-Ask	155	0.606	78/155 (50.3%)
Z	Verified Plan-Execute	155	0.599	86/155 (55.5%)
ZS	Verified PE + Self-Ask	155	0.620	80/155 (51.6%)

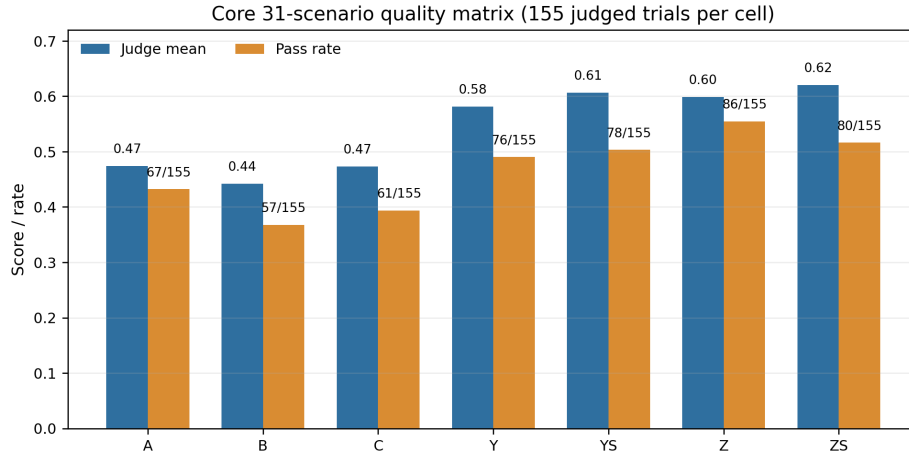


Figure 3: Judge mean and pass rate across the core 31-scenario matrix. The figure highlights that PE-family rows improve judged quality relative to the Agent-as-Tool rows, while Self-Ask is not uniformly positive.

105 The broad matrix supports two main findings. First, MCP standardization does not improve judged
106 quality relative to direct Agent-as-Tool: AT-M is below AT-I on both mean score and pass rate, while
107 optimized MCP recovers mean score but not pass rate. Second, PE-family orchestration improves
108 judged quality relative to the AaT rows. The strongest pass-rate row is Verified Plan-Execute, while
109 the strongest mean-score row is Verified PE + Self-Ask. Self-Ask is therefore a useful ablation but
110 not a uniformly positive lever: it improves Plan-Execute slightly, but lowers the Verified PE pass rate.

5.2 Latency and optimized transport

Table 4: Matched final-six A100 latency cohort. Each row is six scenarios times five trials. Source: results/metrics/gcp_a100_final_matrix_summary.csv.

Cell	Meaning	N	p50 latency	p95 latency	Judge pass
AT-I	Direct tools	30	8.51s	20.73s	6/30
AT-M	MCP baseline	30	12.22s	13.99s	3/30
AT-TP	Optimized MCP	30	12.01s	14.58s	6/30
AT-TPQ	Optimized MCP + serving	30	11.38s	12.87s	3/30
PE-M	Plan-Execute MCP	30	29.62s	60.74s	17/30
V-M	Verified Plan-Execute	30	37.92s	94.93s	22/30
PE-S-TP	PE + Self-Ask + opt. MCP	30	20.26s	41.70s	17/30
V-S-TP	Verified PE + Self-Ask + opt. MCP	30	27.30s	75.22s	16/30

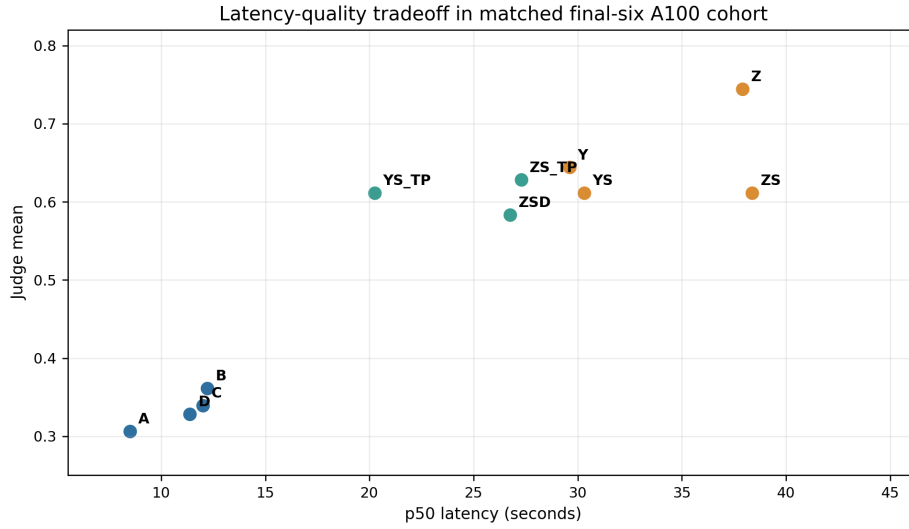


Figure 4: Latency-quality trade-off in the matched final-six A100 cohort. MCP adds p50 latency relative to direct tools; optimized transport improves some latency rows without reliably improving judged quality.

The latency result is also not a simple win story. In the AaT lane, MCP adds single-digit-second p50 overhead relative to direct tools, and optimized MCP transport only slightly reduces that p50 in the matched final-six cohort. The optimized serving stack reduces p95 latency in AaT but still does not improve quality. In the PE-family follow-ons, optimized transport reduces latency substantially relative to the matched PE-family baselines, but this should be read as a systems efficiency result rather than an answer-quality result.

5.3 Hosted-70B top-ups

Hosted-70B rows provide a model-size sanity check rather than a direct replacement for the 8B matrix. Across the post-freeze top-ups, the AaT 70B rows use 45 judged trials per cell and the PE-family 70B rows use 75 judged trials per cell. The strongest hosted-70B mean score is Verified PE + Self-Ask (0.644, 39/75 pass), but the gains are modest relative to the 8B PE-family matrix. We therefore keep 70B as corroborating evidence: larger model calls do not erase the need for grounded tool use and failure accounting.

6 Failure Analysis and Mitigation

Execution cleanliness is not enough for an industrial-agent benchmark. A run that completes without crashing can still be semantically wrong, internally contradictory, or overconfident relative to its own evidence. SmartGridBench therefore records judge scores, failure dimensions, and trajectory-backed failure classifications alongside latency.

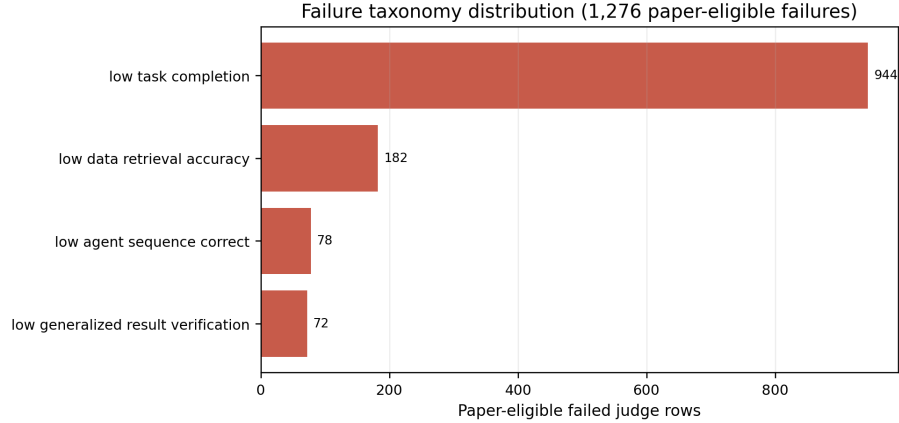


Figure 5: Auto-taxonomy distribution over 1,276 paper-eligible failed judge rows. Source: `results/metrics/failure_taxonomy_current.csv`.

The current failure-taxonomy export contains 1,276 paper-eligible failed judge rows. To calibrate paper examples, we audited a 46-row (`experiment_cell`, `auto_taxonomy_label`) stratified sample against the underlying judge logs and trajectories. This sample is coverage over failure strata rather than a prevalence estimate over all failures. The manual sample confirmed 42 rows, suggested one relabel, and marked three rows as evidence-thin. Among audited rows with a concrete Berkeley-style label, 34 map to task verification and 9 map to inter-agent orchestration; no specification failures appeared in the sample. Recurring patterns include wrong sensor names followed by overconfident outage recommendations, physically incompatible DGA/failure-mode matches, work-order creation without preceding diagnostic evidence, and optimized-serving tool calls with invented failure-mode IDs.

Table 5: Mitigation ladder over the 15-scenario post-freeze cohort. Each row has 75 judged trials. Source: `results/metrics/mitigation_before_after.csv`, rendered from `results/metrics/gcp_post175_mitigation_4tier_summary.csv`.

Lane	Rung	Judge mean	Judge pass	Delta vs. baseline
YS	baseline	0.611	40/75	–
YS	evidence guard	0.476	36/75	-4 pass
YS	repair	0.580	36/75	-4 pass
YS	adjudication	0.524	31/75	-9 pass
ZS	baseline	0.569	36/75	–
ZS	evidence guard	0.509	35/75	-1 pass
ZS	repair	0.578	37/75	+1 pass
ZS	adjudication	0.524	31/75	-5 pass

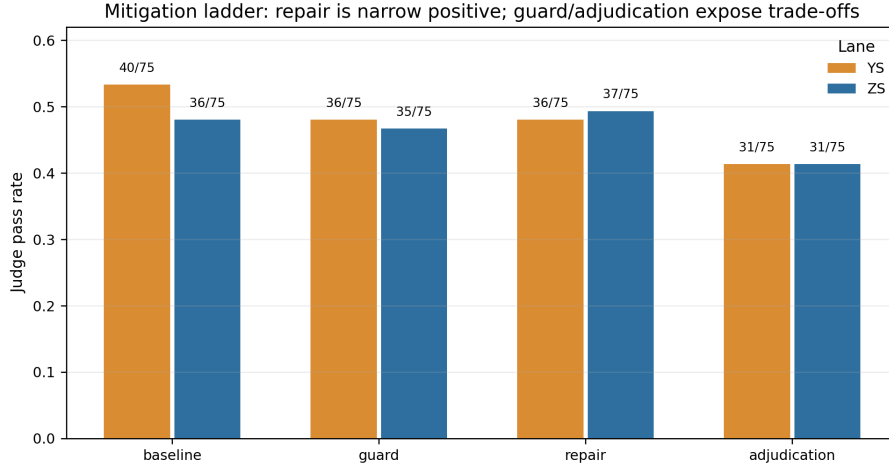


Figure 6: Mitigation ladder pass rates. Repair is the only rung with a positive pass-rate delta, and only narrowly in the verified Self-Ask lane.

140 The mitigation ladder is scientifically useful because it prevents an overly simple universal-lift story.
 141 Detection guardrails and adjudication often make the agent more conservative or expose unresolved
 142 evidence conflicts, which can lower pass rate under the current rubric. Repair is the clearest positive
 143 intervention, but only narrowly in the verified lane. This is exactly the kind of result an evaluations
 144 paper should surface: a benchmark must distinguish error detection, error repair, final-answer quality,
 145 and operational completion.

146 7 Limitations

147 SmartGridBench is a targeted benchmark resource, not a production grid-control system. The
 148 transformer fleet is synthetic and public-safe. Its scenario fixtures are range-checked and operationally
 149 plausible, but the current DGA synthesis does not yet match real DGA distributions under the L3
 150 statistical validator. We therefore avoid claims about real-fleet distributional fidelity.

151 The evaluation is also model- and infrastructure-specific. Most reported rows use Llama-3.1-8B-
 152 Instruct on A100 infrastructure; hosted-70B rows have different sample sizes and should be interpreted
 153 as top-ups. LLM-as-judge scores are useful for repeated evaluation, but they are not a substitute
 154 for full expert review. We mitigate this by preserving judge logs, raw trajectories, manual audit
 155 examples, and failure-taxonomy outputs so readers can inspect the basis for the claims. We also report
 156 repeated-trial counts, means, and pass counts rather than confidence intervals or formal significance
 157 tests; statistical uncertainty estimates are a natural next step once the benchmark stops changing
 158 under the final project deadline.

159 Finally, generated-scenario circularity is a real risk. An LLM can produce domain-shaped text that
 160 passes schema validation while still being redundant or operationally weak. SmartGridBench handles
 161 generated scenarios through staged disposition and bounded promotion, but a larger expert review
 162 pass remains future work.

163 8 Broader Impact

164 Reliable transformer maintenance can reduce outage risk and make industrial AI systems easier
 165 to audit before deployment. The negative side is also clear: over-trusting a tool-using agent in
 166 maintenance operations could accelerate a wrong outage recommendation or hide missing evidence
 167 behind fluent language. SmartGridBench is therefore released as an evaluation and auditing resource,
 168 not as an operational decision system, and the paper emphasizes failure modes and evidence traces
 169 rather than deployment claims.

9 Reproducibility and Release

The submission includes a reviewer-accessible artifact package at <https://huggingface.co/datasets/garn-garn/smartgridbench-review-artifact>. The code package is available at <https://huggingface.co/datasets/garn-garn/smartgridbench-review-artifact/tree/main/code>, and Croissant metadata is available at <https://huggingface.co/datasets/garn-garn/smartgridbench-review-artifact/resolve/main/croissant.json>. The hosted artifact snapshot used for the submission is d606d10765e1ec97e97e3683f6a3794ecf64bb17.

The artifact contains the scenario catalog, canonical scenario JSON files, negative schema fixtures, paper-grade evidence tables, compact judge-score rows, manual judge-audit material, an anonymized executable code package, a file manifest, provenance notes, validation notes, CC BY 4.0 artifact licensing, and MIT code licensing. The repo-level artifact contract preserves run IDs, raw trajectories, judge logs, metrics CSVs, and hardware provenance so table rows can be traced back to concrete executions.

10 Conclusion

SmartGridBench extends AssetOpsBench into Smart Grid transformer maintenance and uses that extension to study protocol, orchestration, and evidence accounting as benchmark variables. The current evidence supports a practical lesson: MCP standardization is measurable and useful, but transport optimization alone does not make an industrial agent reliable. The stronger benchmark signal comes from combining systems measurements with judged quality, manual audit examples, and failure taxonomy so readers can see not only whether an agent ran, but whether it ran fast, grounded its answer, and left enough evidence for a maintainer to trust it.

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